

AI FOOT-PRINT



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ABSTRACT

Artificial Intelligence has become a vital part of modern technology, widely used for text generation, image creation, and data processing. The increasing use of AI systems leads to significant environmental impacts, such as high energy consumption, carbon emissions, and water usage in data centre cooling systems. Despite this, most of the people remain unaware of the hidden environmental cost associated with AI usage.

This project, titled “AI Foot-print’ estimates the environmental impact of AI usage, which aims to create awareness by developing a prototype that estimates the environmental impact of AI usage at individual level. The system takes user input in the form of text and image prompts and dynamically calculates estimated energy consumption, carbon emissions, and water usage using research-based approximation models.

The prototype further classifies user impact into Low, Medium, and High categories to encourage responsible and mindful AI usage. In addition to the prototype development, the project focuses on bridging the gap between technological advancement and environmental responsibility by translating complex AI resource consumption into understandable real-world impact metrics. Features such as impact classification, driving-equivalent comparison, and user awareness feedback were incorporated to help individuals better understand how their AI usage contributes to environmental challenges and to encourage more sustainable digital behaviour.

The findings of this project demonstrate that increased AI usage contributes to higher environmental impact and highlight the importance of sustainable AI awareness and responsible technology consumption.

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INTRODUCTION

Artificial intelligence has become deeply embedded in everyday life, from chatbots and text assistants to image and video generation. To users, these AI systems appear intangible, fast, and almost ‘free’, requiring only a prompt and an internet connection. However, behind the scenes, every AI interaction relies on large scale computing infrastructure, energy supply, and cooling systems that have real environmental consequences. This project titled AI FOOTPRINT aims to reveal and quantify this hidden footprint in simple, understandable way.

AI runs on data centres housing thousands of servers that process model weights, handle user requests, and store vast datasets. These data centres draw electricity thereby contributing to carbon emissions and greenhouse gas. In addition, data centre requires extensive cooling to maintain hardware performance, which often involves use of fresh water.

The environmental cost of AI is not only a technical issue but also a matter of sustainability and social responsibility. As AI usage grows globally, the cumulative effect of small per-user energy and water use can become substantial at the system level. Large language models (LLMs) and image generation models, in particular, are computationally intensive during inference, and their popularity drives higher demand on data centres. Without informed awareness, users may continue to treat AI as a “zero-marginal-cost” service, unaware of the indirect climate and resource impacts [1].

The environmental impact of AI usage indirectly affects human well-being through climate change and resource stress. Higher energy demand from data centres increases carbon emissions, which contributes to global warming, extreme weather events, and rising sea levels that threaten livelihoods, health, and food security. Increased water withdrawals for cooling AI-related infrastructure can strain local water supplies, especially in water-scarce regions, affecting agriculture, drinking water, and ecosystems [2]. Over time, these pressures can lead to more frequent heatwaves, air pollution, and water scarcity, all of which have direct consequences on human health, economic stability, and social equity.

In this context, the present project focuses on building a prototype called the **AI Footprint Calculator**. [3] This tool translates user inputs-specifically the number of text prompts and image prompts-into estimates of energy consumption, carbon emissions, and water usage. The calculator additionally classifies overall impact as low, medium, or high and provides user-friendly feedback, including a driving-equivalent comparison, to help people relate their AI usage to real-world environmental benchmarks. The project explicitly uses research-based approximations and a relative benchmarking methodology rather than claiming absolute precision.

By making the invisible environmental cost of AI visible at the user level, this project seeks to promote more conscious and sustainable digital behaviour. It does not aim to discourage AI usage altogether, but rather to encourage users-especially students, researchers, and professionals-to consider the environmental implications of their interactions and to adopt more responsible AI practices in the long term.

LITERATURE REVIEW

The rapid growth of Artificial Intelligence has prompted increasing research into its environmental impact, particularly in terms of energy consumption, carbon emissions, and water footprint. A growing body of work highlights that while AI services appear lightweight on the user side, they rely on large-scale computing infrastructure whose resource demands are substantial and often underestimated by everyday users.

AI energy consumption and data centres:

Studies on AI and data centre energy use show that computational demand for training and inference of large models can require millions of kilowatt-hours per model. Research on large language models (LLMs) estimates that the energy cost of training a single model can reach around 1.2–1.3 million kWh, corresponding to hundreds of tonnes of CO₂ equivalent emissions. Even the ongoing inference phase-handling user queries-can accumulate significant energy use when scaled across millions of users, turning small per-query consumption into large system-level demand.

A Cornell-led study on the AI data centre boom projects that, by 2030, AI-related computation could emit roughly 24 to 44 million metric tons of CO₂ annually, equivalent to adding several million cars to U.S. roadways. The same study estimates that AI-linked data centres may consume 73 to 11,125 million cubic metres of water per year, comparable to the annual household water use of 6–10 million people [4]. These findings underscore that AI infrastructure is not only energy-intensive but also increasingly water-intensive

Carbon emissions from AI systems:

Research on the carbon footprint of machine learning models has led to specialized tools and frameworks for estimating emissions. The eco2AI project introduces an open-source package that tracks energy consumption and equivalent CO₂ emissions for machine learning models, enabling developers and researchers to monitor and compare the climate impact of different training and inference workflows [5]. Similarly, the LLM Carbon tool proposes an end-to-end model for projecting the carbon footprint of LLMs across training, inference, experimentation, and storage phases, including both operational and embodied carbon.

These studies show that carbon emissions depend strongly on the electricity grid mix, hardware efficiency, and model architecture. For example, models running on grids dominated by coal or natural gas produce higher emissions than those powered by renewable sources. The literature also warns that the current trajectory of AI growth could push the sector's net-zero targets out of reach unless major efficiency gains and faster grid decarbonization are adopted.

Water footprint and data centre cooling:

Water use associated with AI has received less public attention than carbon emissions, but recent work reveals a significant hidden footprint. The study by Li et al., “*Making AI Less ‘Thirsty’: Uncovering and Addressing the Secret Water Footprint of AI Models*,” highlights that data centre cooling and electricity generation for AI systems contribute to large freshwater withdrawals, especially in regions with water-stressed grids [2]. Freshwater is used not only for direct cooling in data centres but also indirectly through power-plant cooling and water-intensive chip manufacturing [6].

Analyses of hyperscale data centres indicate that an average facility can consume around 550,000 gallons of water per day-about 200 million gallons per year-mainly for cooling infrastructure [7]. Much of this water use is geographically concentrated, often in regions already facing water scarcity or competition for freshwater resources. As AI-driven data centres expand, local communities may experience reduced water availability for agriculture, drinking water, and ecosystem services, raising ethical and equity concerns.

Sustainable AI practices and user awareness:

In response to these environmental challenges, researchers and practitioners have begun to advocate for more sustainable AI practices. These include model compression, pruning, quantization, energy-efficient hardware, and workload scheduling tuned to low-carbon grid hours. Platform-level strategies such as “smart siting” of data centres in regions with cleaner grids, faster grid decarbonization, and improved cooling efficiency are also discussed as key levers for reducing AI’s carbon and water impact.

At the user level, several studies emphasize the need for awareness tools that translate AI-usage intensity into understandable environmental units. Existing tools and conceptual frameworks show that benchmarking queries, comparing different models, and exposing per-query carbon and water estimates can help users adjust their behaviour. However, most of these tools remain targeted at developers and researchers, leaving a gap in simple, accessible calculators for general users.

This literature provides the foundation for the present project. The AI Footprint Calculator builds on these insights by using research-based approximations-not precise per-user measurements-to translate everyday AI usage into energy, carbon, and water estimates. Given that exact per-user AI environmental data is not publicly available, the project adopts a **relative** benchmarking methodology that aligns with current studies on AI energy use, carbon emissions, and water footprint, while remaining transparent about its approximate nature.

RESEARCH GAP

Existing research on AI sustainability has made significant progress in quantifying the energy consumption, carbon emissions, and water footprint of machine learning models and data centres. However, most of this work focuses on the infrastructure level, model level, or organisational level, rather than on the individual user level. While tools like eco2AI and LLM Carbon provide detailed carbon-accounting frameworks for developers and researchers, ordinary users generally lack access to simple, intuitive tools that translate their own AI usage into understandable environmental metrics.

Furthermore, per-user environmental data for public AI services is not typically disclosed. Service providers rarely publish exact energy, carbon, or water usage tied to single prompts or sessions, which means that any user-facing calculator must rely on **research-based approximations** and **relative benchmarking** rather than verified per-user figures. The available literature offers rough per-query estimates and model-level benchmarks, but it does not propose a ready-to-use, interactive calculator that ordinary users can employ to build everyday awareness of their digital footprint.

There is also a gap in connecting AI environmental impact to **drivers of behavioural change**. Many studies emphasise the need for sustainable AI practices at the model and infrastructure level, yet few integrate user-facing feedback mechanisms, such as impact classification, driving-equivalent comparisons, or personal usage history, into a single educational prototype. This project addresses that gap by combining existing research on AI energy use, carbon emissions, and water footprint with a **simple, interactive footprint calculator** that raises awareness at the user level.

In summary, the key research gaps addressed by this project are:

1. The absence of accessible, user-oriented tools that translate everyday AI usage into environmental impact estimates;
2. The lack of transparent, approximate calculators that acknowledge the absence of exact per-user environmental data; and
3. The limited integration of sustainable AI concepts into educational, awareness-building interfaces for students, professionals, and general users.

SDG ALIGNMENT

This project explicitly aligns with multiple Sustainable Development Goals (SDGs), as it connects responsible digital behaviour with broader environmental and social objectives. The alignment is both conceptual and practical, as the AI Footprint serves as an educational tool that promotes sustainable consumption, climate-awareness, innovation, and learning.

Primary SDGs:

SDG 12 (Responsible Consumption and Production): The project supports SDG 12 by encouraging responsible use of digital resources. By estimating the energy, carbon, and water footprint of AI usage, the calculator helps users reflect on their consumption patterns and consider more efficient or reduced use of AI where possible. This aligns with the goal of ensuring sustainable consumption patterns in the digital domain, even though the resource use is invisible at the point of interaction.

SDG 13 (Climate Action): By translating AI usage into approximate carbon emissions, the project raises awareness of the climate impact of digital technologies. Users can see how repeated or high-volume AI use contributes indirectly to greenhouse gas emissions, especially when data centres rely on fossil-fuel-based electricity. This awareness supports climate-action initiatives by encouraging individuals and institutions to adopt more sustainable digital practices.

Secondary SDGs:

SDG 9 (Industry, Innovation and Infrastructure): The prototype itself reflects innovation in sustainable digital tools. The AI Footprint Calculator demonstrates how software can be designed not only for functionality but also for environmental transparency and accountability. By promoting energy- and water-aware AI usage, the project indirectly supports the development of more sustainable infrastructure and encourages industries to consider lifecycle environmental impacts when designing AI systems.

SDG 4 (Quality Education): The calculator is educational in nature, helping students, researchers, and professionals understand the environmental side of AI in a practical, hands-on way. By turning abstract concepts such as “carbon footprint” and “water footprint” into concrete numbers and impact levels, the project supports quality education that links technology, environment, and social responsibility. This aligns with SDG 4’s emphasis on education for sustainable development and global citizenship.

OBJECTIVES

The main aim of this project is to design, develop, and evaluate a prototype called the **AI Footprint Calculator**, which estimates the environmental impact of AI usage in terms of energy consumption, carbon emissions, and water usage. The objectives are framed to balance technical implementation with educational and awareness-oriented outcomes, while remaining transparent about the use of research-based approximations and relative benchmarking.

General objectives:

- **To estimate the environmental impact of AI usage** in terms of energy consumption, carbon emissions, and water usage, using research-based approximations and transparent formulas.
- **To build awareness about the “dark side” of AI**, including its dependence on energy-intensive data centres, carbon emissions from electricity use, and hidden water footprint from cooling systems.
- **To encourage sustainable AI usage** by helping users understand how their individual behaviour contributes to environmental pressure and by motivating more responsible digital consumption.

Technical and design objectives:

- To develop a working prototype of the **AI Footprint Calculator** that accepts user inputs in the form of the number of text prompts and image prompts.
- To implement dynamic internal calculations using the following formulas:
 - $\text{Energy (kWh)} = (\text{Text Prompts} \times 0.002) + (\text{Image Prompts} \times 0.02)$, [12]
 - $\text{Carbon Emissions} = \text{Energy} \times 0.5$, [13]
 - $\text{Water Usage} = \text{Energy} \times 1.8$. [11]
- To implement an impact classification logic that categorizes results as:
 - Low ($\text{Energy} < 0.1 \text{ kWh}$),
 - Medium ($0.1 \leq \text{Energy} < 0.5 \text{ kWh}$),
 - High ($\text{Energy} \geq 0.5 \text{ kWh}$).

METHODOLOGY

This project follows a simple input–processing–output approach to estimate the environmental impact of AI usage. Because exact per-user AI environmental data is not publicly available, the calculator uses research-based approximations and relative benchmarking, not precise measurements. [8]

Input–processing–output:

- **Input:** The user enters the number of **text prompts** and **image prompts**.
- **Processing:** The system applies fixed formulas to estimate energy (kWh), carbon emissions, and water usage, then classifies the impact level.
- **Output:** The calculator shows energy, carbon, and water values, an impact class (Low / Medium / High), and a short awareness-based message, including a simple driving-equivalent comparison.

Formulas:

- **Energy (kWh):**

$$\text{Energy} = (\text{Text Prompts} \times 0.002) + (\text{Image Prompts} \times 0.02)$$

- **Carbon emissions:**

$$\text{Carbon Emissions} = \text{Energy} \times 0.5$$

- **Water usage:**

$$\text{Water Usage} = \text{Energy} \times 1.8$$

These formulas are simplified, transparent, and based on average AI-inference energy and emission-factor studies, but they are meant for **awareness**, not exact auditing.

Impact classification and benchmarking:

Impact is classified as:

- **Low:** $\text{Energy} < 0.1 \text{ kWh}$
- **Medium:** $0.1 \leq \text{Energy} < 0.5 \text{ kWh}$
- **High:** $\text{Energy} \geq 0.5 \text{ kWh}$

Because per-user data is not available, the project uses **relative benchmarking**-average values from published AI-footprint studies and calculators-while clearly stating that the outputs are approximate and educational.

PROJECT IMPLEMENTATION

The project implementation involved the development of a working prototype called the *AI Footprint Calculator*, designed to estimate the environmental impact of AI usage at an individual level. The system was developed to calculate estimated energy consumption, carbon emissions, and water usage based on user interaction with AI systems.

The prototype accepts user inputs in the form of the number of text prompts and image prompts. Using predefined research-based approximation formulas, the system dynamically generates environmental impact outputs. The calculations are performed in real-time and the results vary according to the user input values.

The application was designed with a multi-page flow consisting of:

- Input Page
- Results Page
- Impact Classification Page

The system also includes features such as:

- Low/Medium/High impact classification
- Driving-equivalent comparison
- User login and history tracking
- Awareness-focused feedback messages

In addition to the prototype development, dataset analysis and literature-based research were conducted to support the methodology, formulas, and environmental impact estimations used in the project.

RESULTS AND DISCUSSION

The developed prototype successfully generated dynamic environmental impact results based on user input. The system demonstrated that increased AI usage leads to higher energy consumption, carbon emissions, and water usage. It was also observed that image-based prompts consume significantly more resources compared to text-based prompts due to higher computational requirements.

The impact classification feature effectively categorized user activity into Low, Medium, and High levels, helping users better understand their environmental contribution. Features such as driving-equivalent comparison and awareness-based feedback further improved user understanding by converting technical calculations into relatable real-world impact.

The project highlights how AI usage, though often considered virtual, has measurable environmental consequences. Through the prototype, dataset analysis, and research findings, the study emphasizes the importance of sustainable and mindful AI usage practices.

AI usage has a measurable environmental impact. Higher AI usage increases energy, carbon, and water consumption [9]. Increased environmental impact can indirectly affect humans through climate change, resource depletion, and rising energy demand. The AI Footprint Calculator helps create awareness at an individual level. Sustainable and responsible AI usage is necessary for the future.

FUTURE SCOPE

The future scope of this project includes improving calculation accuracy using real-time AI usage data and integrating more advanced environmental tracking systems. The prototype can be further enhanced by collaborating with data centres and AI platforms to obtain more precise energy, carbon, and water consumption metrics.

Integration with real-time AI tracking systems, improved datasets, and collaboration with data centres for more accurate environmental analysis.

Additional features such as real-time monitoring, AI usage optimization suggestions, mobile application integration, and personalized sustainability reports can also be implemented. The project can further be expanded to support multiple AI platforms and larger datasets for more detailed environmental impact analysis.

CONCLUSION

Artificial Intelligence has become an essential part of modern technology, transforming the way people communicate, learn, work, and access information. However, the rapid growth of AI systems has also increased the demand for computational resources, leading to higher energy consumption, carbon emissions, and water usage in data centres and cooling infrastructures. Despite this growing environmental impact, awareness regarding the hidden cost of AI usage remains limited among users. AI can either worsen or support sustainability and that impacts are unevenly distributed [10].

The AI Footprint Calculator developed in this project successfully demonstrates how AI usage can be translated into measurable environmental impact at an individual level. By estimating energy consumption, carbon emissions, and water usage based on user input, the prototype helps users better understand the environmental consequences associated with their AI interactions. The dynamic calculation system, impact classification model, and awareness-based feedback further strengthen the effectiveness of the prototype in promoting responsible AI usage.

The project also highlights how excessive environmental impact can indirectly affect human life through climate change, depletion of natural resources, rising energy demand, and increased pressure on sustainable infrastructure. Through dataset analysis, literature research, and prototype implementation, the study establishes a connection between technological advancement and environmental responsibility.

Overall, this project emphasizes the importance of sustainable and mindful AI usage practices. While the current model is based on research-informed approximations, it provides a strong foundation for future improvements involving real-time tracking systems, industry collaboration, and more accurate environmental datasets. The project ultimately aims to encourage awareness, accountability, and sustainability in the growing field of Artificial Intelligence.

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